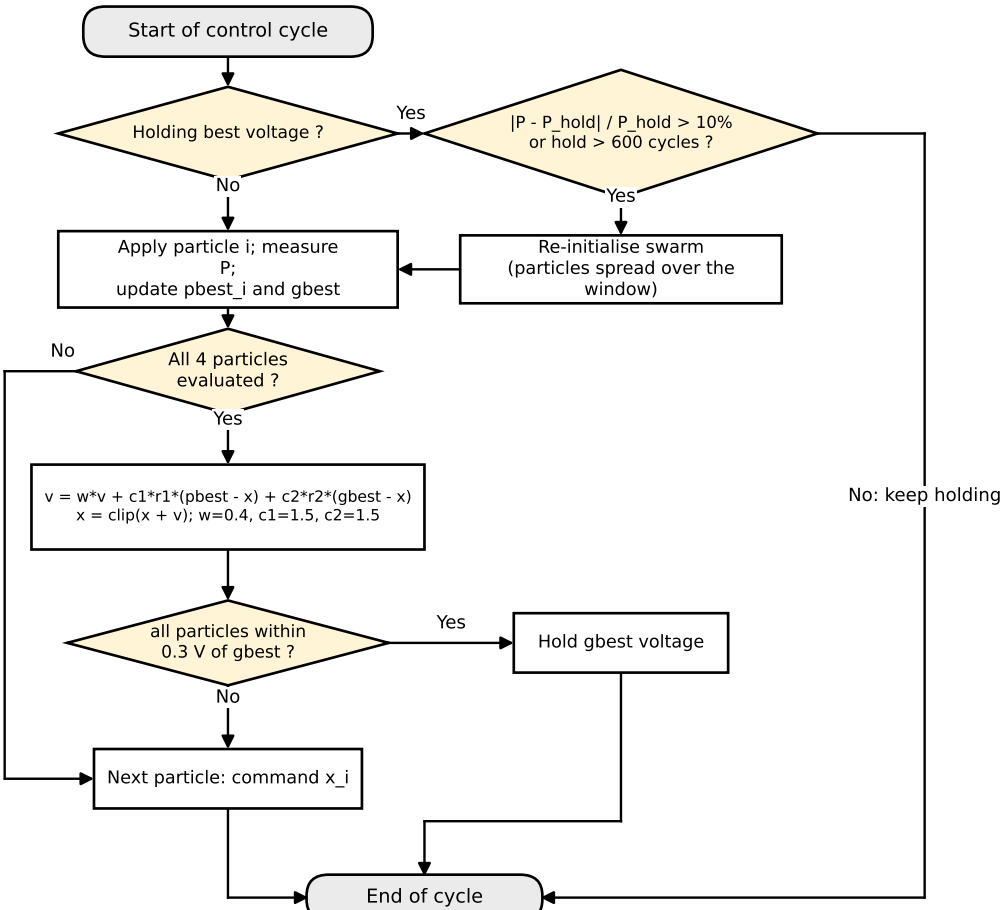
Start of control cycle
Holding best voltage ?
Yes
No
|P - P_hold| / P_hold > 10%
or hold > 600 cycles ?
Yes
No: keep holding
Apply particle i; measure
P;
update pbest_i and gbest
Re-initialise swarm
(particles spread over the
window)
No
All 4 particles
evaluated ?
Yes
v = w*v + c1*r1*(pbest - x) + c2*r2*(gbest - x)
x = clip(x + v); w=0.4, c1=1.5, c2=1.5
all particles within
0.3 V of gbest ?
Yes
Hold gbest voltage
No
Next particle: command x_i
End of cycle